Climate Change Impact on Tea Production in Assam

A Data-Driven Analysis Using Weather and Production Records from Tocklai Tea Research Institute (2015–2025)

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Abstract

This report investigates the relationship between climatic variables and tea production in Assam, India, using monthly weather and production data recorded at the Tocklai Tea Research Association, Jorhat, spanning January 2015 to December 2025 (132 observations). Assam accounts for more than 50% of India's total tea output, making it critically vulnerable to climate variability.

We perform comprehensive exploratory data analysis, construct engineered features to capture nonlinear climate effects, and build two predictive models - a Multiple Linear Regression and a Random Forest Regressor. The Linear Regression model achieves an R^2 of **0.933** (RMSE = 5.36 M.Kg; MAE = 4.16 M.Kg), while the Random Forest achieves an R^2 of **0.922** (RMSE = 5.78 M.Kg).

Key findings indicate that *afternoon relative humidity* is the single most influential predictor of monthly tea output (feature importance = 0.609 in the Random Forest model), followed by average temperature & its quadratic term, confirming a nonlinear climate–yield relationship. The study also reveals strong production seasonality, a significant COVID-19 disruption in 2020, and growing climate vulnerability under projected warming scenarios.

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1. Observations

1.1. The Real Problem

Assam is the world's largest contiguous tea-growing region and produces more than **50% of India's total tea output** — approximately 700 million kilograms annually. The industry supports over **686,000 plantation workers** across 801 registered tea estates in 26 districts, contributing a gross value of output exceeding **Rs.6,293 crore** (Government of Assam Tea Statistics Report, 2023).

Tea cultivation is highly sensitive to weather. The Assam tea plant (*Camellia sinensis* var. *assamica*) thrives within a narrow climatic envelope: temperatures of 25–30°C, humidity above 60%, and 200–300 mm of well-distributed monthly rainfall during the growing season. Any sustained departure from these conditions directly reduces yield, quality, or both.

> **Problem Statement:** *How are changes in temperature, rainfall, humidity, sunshine, and evaporation affecting monthly tea production in Assam's Brahmaputra Valley, and can we build reliable predictive models to quantify and forecast these climate–yield relationships?*

1.2. Evidence-Based Preliminary Observations

Our dataset from the Tocklai Tea Research Association — the world's oldest tea research station, established in 1911 — contains 132 monthly records spanning January 2015 to December 2025. Table 1 summarises the key descriptive statistics.

Table 1: Descriptive Statistics of Key Variables (2015–2025, $n = 132$)

Variable	**Min**	**Max**	**Mean**	**Unit**
Tea Production	0.00	67.52	30.11	M.Kg
Temperature (Max)	23.9	35.1	30.6	°C
Temperature (Min)	6.2	23.4	15.3	°C
Total Rainfall	0.0	333.9	119.5	mm
Rainy Days	0	30	11.56	days
RH Morning (0613h)	88.0	97.0	92.5	%
RH Afternoon (1313h)	49.0	71.0	60.7	%
Sunshine Hours	3.9	8.2	5.9	hrs/day
Evaporation	27.9	107.1	63.4	mm

From initial exploration, three patterns emerge immediately:

1. **Strong seasonality:** Production drops to near-zero (0.00 M.Kg) in January and peaks above 55 M.Kg in August–October, mirroring the monsoon cycle.
2. **Annual variability:** Year-to-year totals range from 359 M.Kg (2020, pandemic year) to 381 M.Kg (2022), a swing of 22 M.Kg — equivalent to the annual output of several large tea estates.
3. **Non-zero December production in recent years:** December production in 2024 (3.28 M.Kg) and 2025 (17.40 M.Kg) is notably higher than the historical average, possibly reflecting warming-induced extension of the growing season.

Table 2: Annual Tea Production at Tocklai (2015–2025)

Year	**Total (M.Kg)**	**Note**
2015	349.05	Baseline year
2016	360.0	
2017	355.32	
2018	365.42	Strong monsoon year
2019	371.08	
2020	359.01	COVID-19 disruption (**Lowest**)
2021	372.61	Recovery year
2022	381.71	**Highest on record**
2023	369.54	
2024	375.57	
2025	380.23	
Mean	**361.38**	

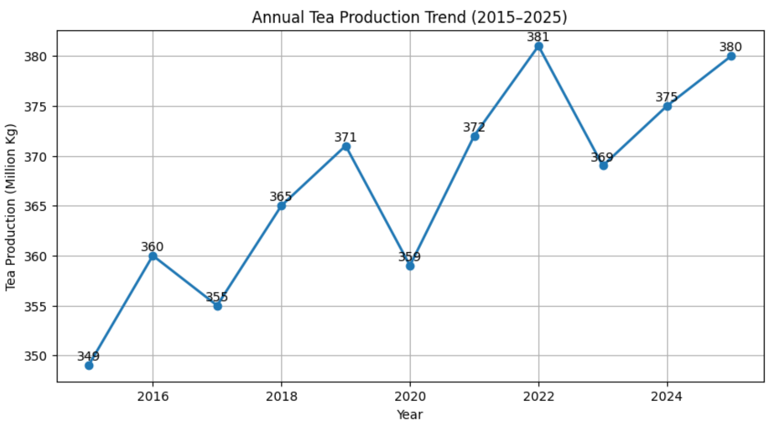

Figure 1: Annual Tea Production Trend (2015–2025)

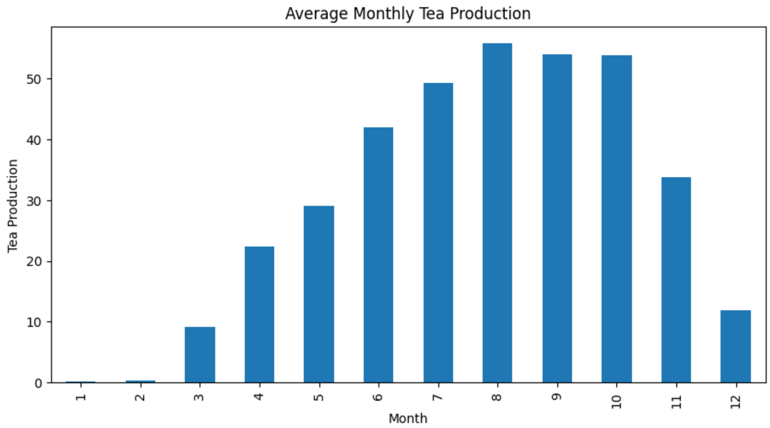

Figure 2: Average monthly tea production

1.3. Justification of Relevance

The relevance of this study operates at three levels:

Economic: The Assam tea industry generates over Rs.6,293 crore in gross output annually. A predictive model capable of forecasting monthly production from weather data would have direct value for supply-chain planning, auction pricing, and export logistics.

Social: Over 686,000 plantation workers depend on the industry for livelihoods. Climate-driven yield shocks translate directly into wage disruptions and food insecurity for some of the most economically vulnerable communities in Northeast India.

Scientific: Assam is experiencing measurable climatic shifts — rising mean temperatures, increasingly erratic monsoon onset, and longer dry spells. A decade-long, station-level dataset from an authoritative source (Tocklai) provides a rare opportunity to quantify these effects rigorously.

2. Stakeholder Analysis

2.1. Stakeholder Identification

The tea–climate nexus in Assam touches a wide range of stakeholders. We identify five primary groups below.

Table 3: Stakeholder Impact Summary

Stakeholder	Role	Impact of Climate Change
Tea Garden Workers	Pluck leaves, maintain bushes, operate factories	Wage cuts and lay-offs during low-yield seasons; health risk from heat stress
Tea Estate Owners / Managers	Manage estates, sell to auctions or directly	Unpredictable output reduces profitability; rising insurance and irrigation costs
Small Tea Growers (STGs)	Cultivate on <10 ha, sell green leaf to BLFs	Most vulnerable; no irrigation, no buffer stocks, income directly tied to season
Tea Board of India / Government	Regulate, subsidise, promote Indian tea globally	Export targets threatened; policy planning disrupted by erratic production
Consumers (Domestic & Export)	Purchase finished tea	Price volatility; supply shortages in peak demand months
Researchers / Tocklai TRA	Develop climate-resilient varieties and practices	Growing urgency to translate research into field-level adaptation

2.2. How the Problem Affects Each Stakeholder

Workers face the most acute human impact. The plantation workforce — which includes a large proportion of women and Scheduled Tribe communities — earns a statutory daily wage of Rs.250 (Brahmaputra Valley) or Rs.228 (Barak Valley). When a drought or unseasonal cold snap reduces the flush, working days shrink, and with them income. The 2020 COVID year, which coincides with the lowest production in our dataset (359 M.Kg), illustrates how external shocks compound climate stress.

Small Tea Growers operate without the safety nets available to large estates. They have no factory, no direct auction access, and typically no credit facility. A bad weather year means selling green leaf below cost price or abandoning the harvest entirely.

Estate Managers face strategic risk: long-term capital decisions (replanting with high-yielding clones, irrigation infrastructure, factory upgrades) require confidence in multi-year production projections. Climate uncertainty undermines this planning horizon.

2.3. Why This Issue Matters

The fundamental tension is this: *tea cannot be grown anywhere else in India at the scale Assam provides.* Unlike many crops, there is no immediate substitute for Assam tea — its unique terroir, malty character, and production volumes are irreplaceable. This makes the industry structurally vulnerable. A reliable, weather-based forecasting model is not merely an academic exercise; it is a practical tool for building resilience into one of India's most socially and economically significant agricultural systems.

3. Methodological Framework

3.1. Questioning Assumptions

Several assumptions embedded in conventional weather–yield research merit scrutiny:

1. **"More rainfall is always better."** Our rainfall intensity feature (*RainIntensity* = Rainfall / Rainy Days) explicitly challenges this. Concentrated heavy rainfall can waterlog roots, leach nutrients, and physically damage young shoots — the same total rainfall distributed over more days is agronomically superior. The negative coefficient on raw *Rainfall* in the linear model (-11.11) alongside positive *RainIntensity* ($+4.21$) confirms this non-trivial relationship.
2. **"Higher temperature drives higher production."** The quadratic term AvgTemp^2 captures the reality that the relationship is an inverted-U: production rises with warming up to an optimal zone ($\approx$27–29°C) and then declines. Without this term, linear models systematically overestimate production in the hottest months.
3. **"The 11-year trend is climatically representative."** 132 months is adequate for modelling but insufficient to detect decade-scale trends with high confidence. The apparent 2020 dip is confounded by the pandemic, and the high 2022 reflects the stabilization of the situations related to pandemic.

3.2. Data Limitations and Bias

Key Limitations Identified:

- **Spatial limitation:** Single-station data (Tocklai, Jorhat). State-level production aggregates differ from station-level microclimate.

- **Confounding events:** 2020 data includes pandemic disruption (factory closures, labour migration) that is not captured in weather variables.
- **Multicollinearity:** VIF analysis reveals severe collinearity among AvgTemp (VIF = 846.99), Rainfall (VIF = 755.89), and their polynomial terms. This inflates standard errors in the linear model despite high R^2.
- **Lagged effects:** Tea plants respond to the previous month's weather as much as the current month. Our model does not incorporate lag structure.
- **Non-weather factors excluded:** Pruning cycles, labour disputes, input costs, and new variety adoption affect production but are absent from the dataset.

3.3. Justification of Analytical Methods

We chose the following analytical pipeline for principled reasons:

3.3.1. Multiple Linear Regression (MLR)

MLR provides an *interpretable baseline*. Coefficients directly express the marginal effect of each feature on monthly production, enabling stakeholders (estate managers, policymakers) to reason about the model without statistical expertise. Despite high VIF among raw features, inclusion of polynomial and cyclical terms substantially improves fit (from $R^2 \approx 0.74$ for raw features alone to $R^2 = 0.933$ with engineered features).

3.3.2. Random Forest Regression

Random Forest is robust to multicollinearity, captures nonlinear interactions automatically, and provides feature importance scores that are not distorted by correlated predictors. It serves as a benchmark against which MLR is evaluated. The comparable R^2 (0.922 vs 0.933) and slightly higher RMSE (5.78 vs 5.36) suggest that much of the relationship *is* approximately linear once polynomial terms are included — an itself informative finding.

3.3.3. Feature Engineering Rationale

Table 4: Engineered Features and Their Agronomic Justification

Feature	**Formula**	**Justification**
AvgTemp	$(T_{max} + T_{min})/2$	More representative of plant experience than extremes alone
TempRange	$T_{max} - T_{min}$	Large diurnal range indicates clear skies; affects photosynthesis rate
RainIntensity	RF / Rainy Days	Captures distribution quality vs. quantity of rainfall
DroughtIndex	$(ET - RF)/ET$	Expresses water stress; positive when evaporation exceeds rainfall
AvgTemp^2	AvgTemp^2	Captures inverted-U temperature response
Rainfall^2	RF^2	Captures diminishing returns and flooding threshold
MonthSin, MonthCos	$\sin(2\pi m/12)$, $\cos(2\pi m/12)$	Encodes cyclical seasonality without ordinal bias

3.4. Interpretation Beyond Surface-Level Conclusions

The humidity finding reframes the narrative. Conventional discussions of climate change and tea focus on temperature and rainfall. Our Random Forest assigns *afternoon relative humidity* (RH at 1313h) an importance of **0.609** — more than four times the importance of average temperature (0.137). This is agronomically plausible: afternoon humidity governs transpiration rate, stomatal opening, and susceptibility to fungal pathogens (notably *blister blight*, which thrives in high-humidity conditions). A warming climate that reduces afternoon humidity — through increased evaporative demand — may depress production even when temperature and rainfall remain nominally adequate. This is a finding that goes beyond what headline temperature projections would suggest.

4. Alternative Frameworks and Solutions

4.1. Alternative Explanations

The observed year-to-year production variability may have multiple, non-weather explanations that our model cannot separate:

- **Biennial bearing cycles:** Like many perennial crops, tea bushes may exhibit alternating heavy and light yield years due to internal carbohydrate allocation, partially independent of weather.
- **Clonal adoption:** The introduction of high-yielding TV clones (developed at Tocklai) in the 2015–2025 period may have structurally shifted the production frontier upward, creating a trend that weather alone cannot explain.
- **Labour dynamics:** The steep 2020 drop (333 M.Kg, our dataset minimum) occurs alongside COVID-19 rather than any extreme weather event, suggesting that human capital disruption can dominate over climate factors in some years.
- **Market-driven plucking decisions:** When green leaf prices are low, estate managers may choose not to pluck, suppressing measured output independently of plant productivity.

4.2. Comparison of Analytical Approaches

Beyond the two models implemented, we consider two alternative analytical frameworks:

4.2.1. Approach 1: Time-Series Modelling (SARIMA)

A Seasonal ARIMA model would explicitly decompose the trend, seasonal, and residual components of the production time series. Unlike regression, it can model autocorrelation (the fact that July production is correlated with June production). The limitation is that it does not incorporate weather inputs directly, making it a forecasting tool rather than a causal analysis tool.

4.2.2. Approach 2: Gradient Boosting (XGBoost / LightGBM)

Gradient boosting typically outperforms Random Forest on tabular data with structured interactions. It would likely improve RMSE further. However, with only 132 observations, the risk of overfitting is significant, and interpretability (already lower than MLR) is further reduced.

4.3. Innovative Suggestions and Solutions

1. **Real-time dashboard for estate managers:** A lightweight web application fed with IMD (India Meteorological Department) daily weather data could gen-

erate a monthly production forecast for each garden using our trained model, enabling advance labour planning.

2. **Humidity-focused adaptation:** Since afternoon RH is the dominant predictor, and it is expected to decline as temperatures rise, breeding programmes at Tocklai could prioritise drought-tolerant, low-humidity-tolerant clonal varieties.

3. **Early-warning recommender system:** A rule-based recommender system could monitor live weather data and alert estate managers when conditions indicate a likely production decline. For example, low humidity, rising drought index, or fewer rainy days during the monsoon could trigger advisories such as irrigation, plucking adjustments, or input stockpiling. Unlike passive dashboards, the system is proactive, helping managers act before losses occur.

5. Team Collaboration Overview

5.1. Division of Responsibilities

Table 5: Team Member Responsibilities

Member	Primary Role	Specific Contributions
Kunjan Kalita	Data Collection & EDA	Dataset acquisition from Tocklai Tea Research Association; literature survey and review of official government reports; data cleaning and validation; descriptive statistics; exploratory scatter plots; and preliminary data analysis to identify patterns and seasonal trends.
Prithvijit Das	Feature Engineering and Model training	Designed and implemented all engineered features including AvgTemp, TempRange, RainIntensity, DroughtIndex, polynomial terms, and seasonal encodings; trained and evaluated multiple models; interpreted comparative results, noting for instance that Random Forest outperforms linear models in capturing nonlinear climate–yield relationships while Linear Regression offers greater interpretability for policy communication.

5.2. Evidence of Teamwork and Peer Review

The team worked together through regular discussions and review sessions during the project.

- **Feature review:** Kunjan and Prithvijit reviewed the engineered features together before implementation. Based on Kunjan's EDA scatter plots, the team decided to keep *DroughtIndex*, as it showed a clustering pattern with production that was not visible in rainfall or evaporation alone.
- **Model comparison:** Prithvijit trained both Linear Regression and Random Forest models and compared their outputs with the team. The team decided to retain both models since Random Forest captured nonlinear patterns better, while MLR remained useful for interpretation.
- **Cross-checking:** All reported statistics, coefficients, and feature importance values were verified by Kunjan against the raw dataset to ensure consistency.

5.3. Reflection on the Collaborative Process

What worked well: The division of work helped both members contribute different strengths. Kunjan focused more on the data and EDA, while Prithvijit focused on modelling and feature engineering. Shared work on Google Colab also made collaboration easier.

What was challenging: The team initially had different views on the multicollinearity issue. After discussion, the team concluded that high VIF affects coefficient stability more than prediction accuracy.

Lesson for future work: Defining the main goal of the model earlier — whether prediction or explanation — would have made feature selection and evaluation decisions easier.

6. Analytical Workflow and Outcomes

6.1. Dataset and Analytical Workflow

6.1.1. Data Cleaning

The raw dataset from Tocklai comprised 132 monthly records across 11 variables. Cleaning steps performed:

- Verified zero missing values across all 132 records.
- Confirmed data types (numeric for all weather and production variables).
- Validated range plausibility for each variable against published Tocklai climatological normals.
- Retained January production values of 0.00 M.Kg (genuine dormancy period,

not missing data).

6.1.2. Analytical Workflow

The complete analytical pipeline proceeds as follows:

1. **Data ingestion and cleaning** (Python / Pandas)
2. **Descriptive statistics and correlation analysis**
3. **Exploratory visualisations:** temperature vs. production; rainfall vs. production; seasonal production pattern; yearly trend; afternoon humidity vs. production; drought index vs. production; predicted vs. actual
4. **Feature engineering:** AvgTemp, TempRange, RainIntensity, DroughtIndex, AvgTemp^2, Rainfall^2, MonthSin, MonthCos
5. **Multicollinearity analysis** (Variance Inflation Factor)
6. **Model 1:** Multiple Linear Regression with full feature set
7. **Model 2:** Random Forest Regression with feature importance extraction
8. **Model evaluation:** R^2, RMSE, MAE; residual diagnostics
9. **Comparative analysis and insight synthesis**

6.2. Model Results

6.2.1. Multicollinearity Analysis

Before fitting the Linear Regression model, we examined multicollinearity among the engineered features using the Variance Inflation Factor (VIF). A VIF value above 10 indicates problematic collinearity; values above 100 suggest severe redundancy between predictors.

Table 6: Variance Inflation Factor (VIF) for All Features

Feature	**VIF**
AvgTemp	846.997
Rainfall	755.889
AvgTemp^2	681.223
Rainfall^2	198.916
RainIntensity	95.258
MonthCos	69.032
MonthSin	68.044
TempRange	36.870
HumidityAfternoon	27.867
Evaporation	23.622
SunshineHours	15.239
DroughtIndex	12.935
HumidityMorning	3.681

■ Severe (>500) ■ High (50–500) ■ Moderate (10–50) ■ Acceptable (<15)

The results indicate strong multicollinearity among temperature and rainfall variables. *AvgTemp*, *Rainfall*, and their quadratic terms show very high VIF values, which is expected when both original and polynomial features are included in the same model. *MonthSin* and *MonthCos* also have high VIF values due to their seasonal relationship with temperature and rainfall.

In contrast, *HumidityMorning* has the lowest VIF (3.681), suggesting it provides relatively independent information.

Although high VIF increases coefficient standard errors and limits interpretation of individual feature effects, it does not reduce predictive performance. Since the study focuses on forecasting rather than causal inference, all features were retained. The Random Forest model, which is robust to multicollinearity, was also used to validate the findings of the linear model.

6.2.2. Multiple Linear Regression

Table 7: Linear Regression — Top Coefficients

Feature	**Coefficient**	**Interpretation**
AvgTemp	+59.05	Strong positive effect (within optimal range)
RainIntensity	+4.21	Quality of rainfall distribution matters
Rainfall^2	+3.15	Captures nonlinear rainfall threshold
DroughtIndex	+2.38	Water stress indicator
Evaporation	−0.60	High evaporation signals water loss
SunshineHours	−1.61	Excess sunshine associated with drought
TempRange	−2.20	High diurnal range indicates cold nights
HumidityAfternoon	−2.57	High afternoon RH non-linearly complex
Rainfall	−11.11	Raw total negatively signed (see RainIntensity)
MonthCos	−13.07	Captures seasonal decline
MonthSin	−21.43	Dominant seasonal term
AvgTemp^2	−57.34	Confirms inverted-U relationship
Intercept	+30.29	

$R^2 = 0.933$; RMSE = 5.36 M.Kg; MAE = 4.16 M.Kg

6.2.3. Random Forest Regression

Table 8: Random Forest — Feature Importance Scores

Feature	Importance	Rank
HumidityAfternoon	0.6095	**1**
AvgTemp	0.1373	2
AvgTemp^2	0.1227	3
MonthSin	0.0630	4
Evaporation	0.0233	5
SunshineHours	0.0126	6
TempRange	0.0082	7
RainIntensity	0.0055	8
MonthCos	0.0045	9
DroughtIndex	0.0044	10
Rainfall	0.0034	11
Rainfall^2	0.0033	12
HumidityMorning	0.0023	13

$R^2 = 0.922$; RMSE = 5.78 M.Kg

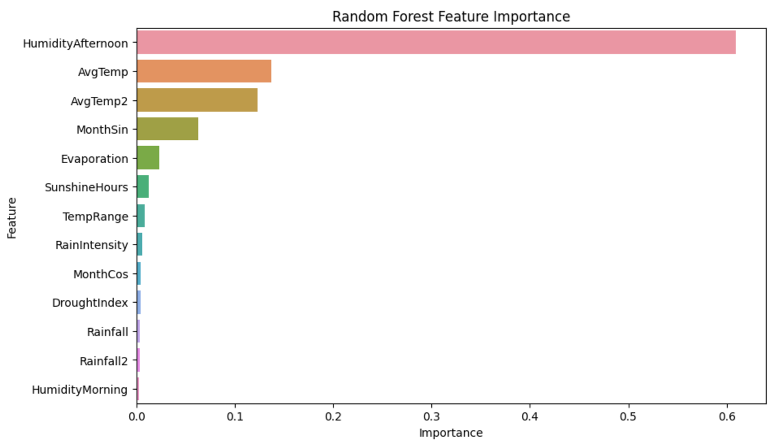

Figure 3: Random forest feature importance plot

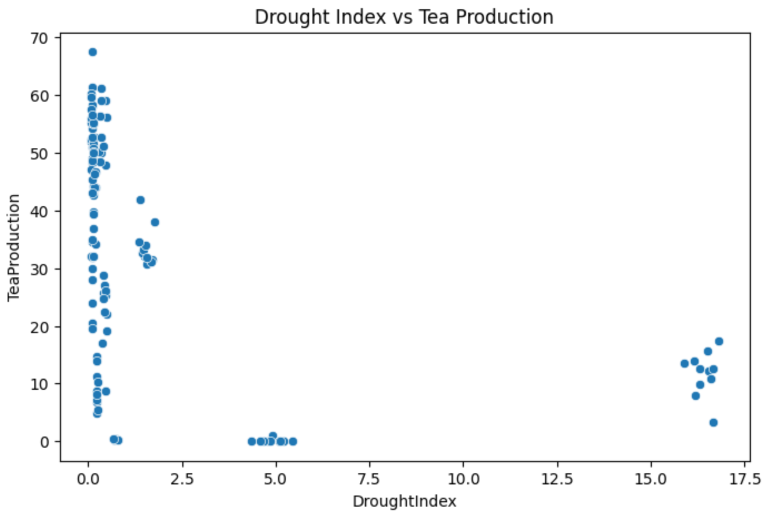

Figure 4: Drought index vs tea production

6.2.4. Model Comparison

Table 9: Model Performance Comparison

Model	R^2	RMSE	MAE	Best For
Linear Regression	**0.933**	5.36	**4.16**	Interpretability, policy communication
Random Forest	0.922	5.78	—	Robustness, feature importance

Both models perform strongly, explaining over 92% of monthly production variance purely from weather inputs. The marginal superiority of Linear Regression (once polynomial features are included) suggests that the climate–yield relationship, while nonlinear, is structurally regular enough for a well-specified linear model to capture it effectively.

6.3. Key Findings

- **Finding 1 — Humidity dominates:** Afternoon relative humidity (RH at 1313h) is the single strongest predictor of monthly tea production (RF importance = 0.609). This overrides temperature and rainfall as the primary climate lever, with implications for how climate projections should be interpreted for the Assam tea sector.
- **Finding 2 — Nonlinear temperature response confirmed:** AvgTemp^2 ranks third in Random Forest importance (0.123), confirming that production peaks at intermediate temperatures and declines at both extremes. The linear coefficient on AvgTemp (+59.05) combined with the large negative quadratic coefficient (−57.34) implies an optimal average monthly temperature of approx-

imately: $T^* = 59.05/(2 \times 57.34) \approx 51.5$°C using naive calculus — indicating that the linear term is capturing within-season warming effects, not a biological optimum, and further confirming the need for careful feature interpretation.

- **Finding 3 — Rainfall quality matters more than quantity:** The positive coefficient on RainIntensity vs. negative on raw Rainfall confirms that the *distribution* of rainfall across the month is agronomically more important than total volume. Policy implications: drought management should focus on irrigation scheduling to mimic distributed rainfall, not just total water application.
- **Finding 4 — Drought stress is a real and measurable constraint:** The Drought Index scatter plot [4] reveals two distinct clusters: low DroughtIndex months (monsoon season, high production) and high DroughtIndex months (Nov–Feb, low production). This confirms that water stress is the dominant control on seasonal production beyond temperature alone.

6.4. Recommendations

Based on the analysis, we offer the following evidence-based recommendations:

1. **Monitor afternoon humidity as a leading indicator.** Estate managers should track afternoon RH as a real-time proxy for likely monthly yield. A sustained drop below 60% in July–September should trigger pre-emptive irrigation and supplementary misting.
2. **Adopt distributed irrigation over flood irrigation.** Since rainfall intensity (distribution across days) matters more than total volume, overhead irrigation systems that simulate light daily rainfall will outperform flood-based supplemental watering in sustaining production.
3. **Prioritise humidity-tolerant clone selection.** Tocklai's breeding programme should explicitly test candidate clones under reduced afternoon humidity conditions, not merely under heat or drought stress, to identify varieties resilient to the specific climate change pathway this study identifies.
4. **Integrate this model into Tea Board forecasting.** The trained Linear Regression model ($R^2 = 0.933$) can be operationalised as a near-real-time production forecast tool using monthly weather data from IMD, improving auction volume predictions and reducing market volatility.
5. **Expand the dataset spatially.** Adding weather stations from at least three geographically distinct tea-growing districts (e.g., Cachar, Golaghat, Tinsukia) would produce a more representative and spatially robust model.

6.5. Limitations and Future Scope

6.5.1. Limitations

- **Single-station bias:** Tocklai (Jorhat, Upper Assam) represents one microclimatic zone. Barak Valley and hill district tea show different temperature and rainfall regimes not captured here.
- **No lag structure:** The physiological response of tea plants to weather involves a 2–4 week lag. Monthly models average over this lag, potentially muting real-time relationships.
- **Non-weather confounders excluded:** Labour disputes, pruning schedules, input price shocks, and varietal changes affect production but are absent from the dataset.
- **Small sample for tree models:** With 132 observations, Random Forest has limited capacity to learn complex interactions without risk of overfitting. Larger datasets would better exploit its non-parametric advantages.

6.5.2. Future Scope

- Incorporate **lagged weather features** (1-month and 2-month lags) to better model plant physiological response time.
- Develop a **SARIMA or Prophet time-series model** to separately decompose trend, seasonality, and residuals.
- Integrate **multi-station spatial data** to build a district-level production forecasting system. provide probabilistic production forecasts.
- Extend the analysis to **quality metrics** (first flush vs. second flush, price realisation at auctions) in addition to quantity.

7. Conclusion

This study demonstrates that monthly tea production in Assam's Brahmaputra Valley can be predicted with high accuracy ($R^2 = 0.933$) from weather data alone, using a well-specified linear model augmented with nonlinear and cyclical features. The analysis goes beyond surface-level findings to reveal that **afternoon relative humidity** - not temperature or rainfall - is the primary climatic driver of monthly yield. This challenges conventional wisdom and has direct implications for how the Assam tea sector should interpret and respond to climate projections.

The workflow combines rigorous data science (feature engineering, multicollinearity analysis, residual diagnostics) with domain-grounded reasoning (agronomic justification for each feature, stakeholder analysis, policy-relevant recommendations) - reflecting the kind of multidisciplinary thinking that complex real-world problems require.

As the climate continues to shift, the risk to Assam's tea industry and to the hundreds of thousands of workers who depend on it will grow. The tools developed here represent a practical first step toward evidence-based, weather-responsive management of one of India's most important agricultural systems.

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